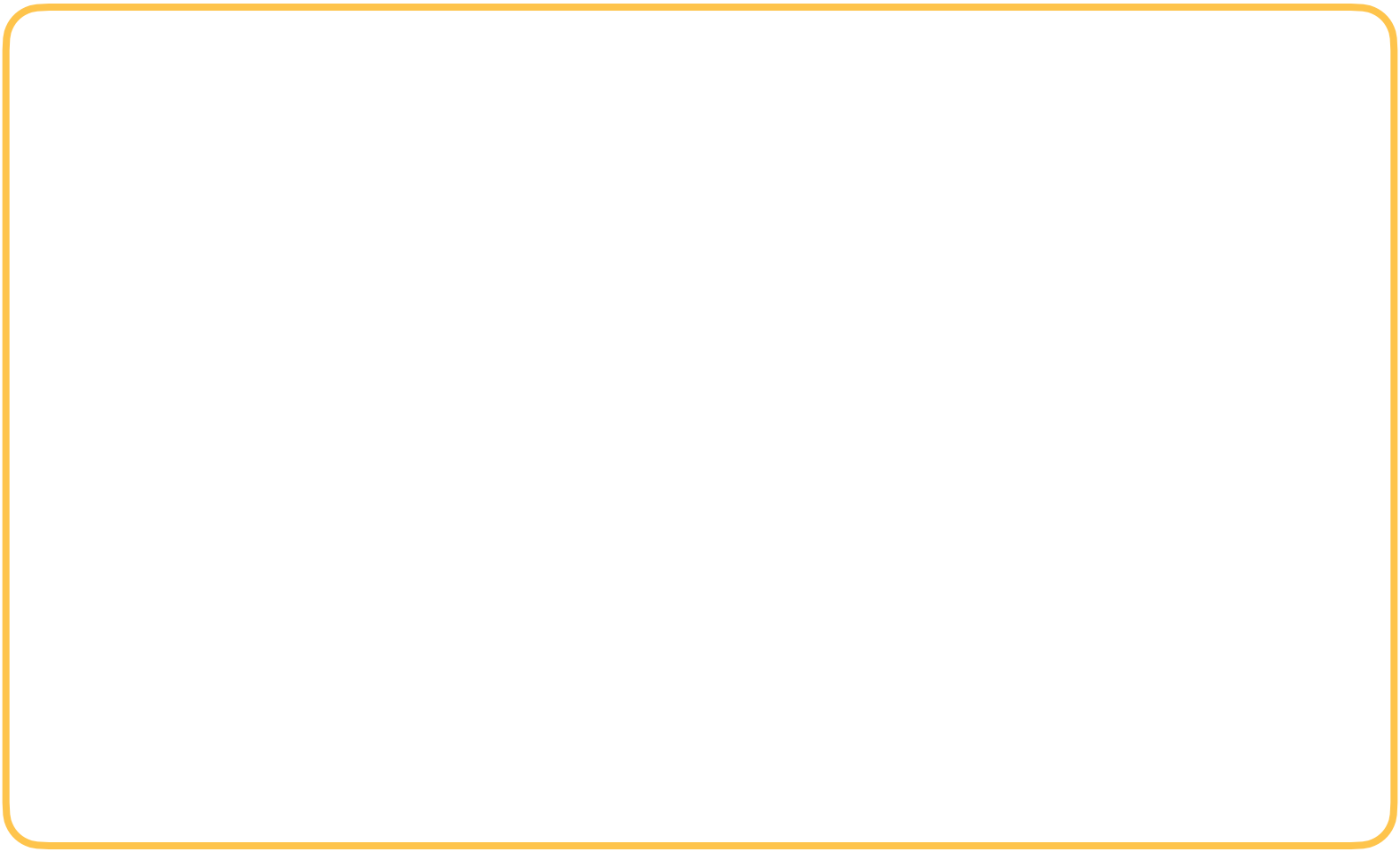








Three Canonical Energies

Three types of “nice” functionals often used in Wasserstein grad flows:

Internal Energy

$$\mathcal{F}(\rho) = \int_{\mathbb{R}^d} f(\rho(x)) \, dx$$

Intuition: penalizes local density (entropy, porous medium, etc.).

$$\text{First var.: } \frac{\delta \mathcal{F}}{\delta \rho}(x) = f'(\rho(x))$$

Potential Energy

$$\mathcal{V}(\rho) = \int_{\mathbb{R}^d} V(x) \rho(x) \, dx$$

Intuition: external field / drift (confinement, data term).

$$\text{First var.: } \frac{\delta \mathcal{V}}{\delta \rho}(x) = V(x)$$

Interaction Energy

$$\mathcal{W}(\rho) = \frac{1}{2} \iint W(x - y) \rho(x) \rho(y) \, dx dy$$

Intuition: nonlocal attraction/repulsion.

$$\text{First var.: } \frac{\delta \mathcal{W}}{\delta \rho}(x) = (W * \rho)(x)$$

$$\text{Total energy (additive model): } \mathcal{E}(\rho) = \underbrace{\mathcal{F}(\rho)}_{\text{internal}} + \underbrace{\mathcal{V}(\rho)}_{\text{potential}} + \underbrace{\mathcal{W}(\rho)}_{\text{interaction}}$$

$$\text{Resulting Wasserstein gradient flow (PDE): } \partial_t \rho = \nabla \cdot \left(\rho \nabla [f'(\rho) + V + (W * \rho)] \right)$$

Further Reading

{Euclidean, Metric, and Wasserstein} Gradient Flows: an overview

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Abstract

This is an expository paper on the theory of gradient flows, and in particular of those PDEs which can be interpreted as gradient flows for the Wasserstein metric on the space of probability measures (a distance induced by optimal transport). The starting point is the Euclidean theory, and then its generalization to metric spaces, according to the work of Ambrosio, Gigli and Savaré. Then comes an independent exposition of the Wasserstein theory, with a short introduction to the optimal transport tools that are needed and to the notion of geodesic convexity, followed by a precise description of the Jordan-Kinderlehrer-Otto scheme, with proof of convergence in the easiest case: the linear Fokker-Planck equation. A discussion of other gradient flows PDEs and of numerical methods based on these ideas is also provided. The paper ends with a new, theoretical, development, due to Ambrosio, Gigli, Savaré, Kuwada and Ohta: the study of the heat flow in metric measure spaces.

